

# NING WANG

Amherst, MA | +1 (413) 275-6990 | [ninwang@umass.edu](mailto:ninwang@umass.edu)

[Personal website](#) | [GitHub](#) | [Google Scholar](#)

ECE Ph.D. researcher building Python/PyTorch learning systems and network simulations. Seeking Summer 2027 software, machine learning, and network systems internships.

## EDUCATION

### University of Massachusetts Amherst

Ph.D., Electrical and Computer Engineering (in progress) | Feb 2024 - Present

M.S., Electrical and Computer Engineering | Jan 2024 | GPA: 3.914/4.0

### University of Electronic Science and Technology of China

B.S., Communication Engineering | Jul 2022 | GPA: 3.50/4.0

## TECHNICAL SKILLS

Programming: Python, MATLAB, C, C++ | ML: PyTorch | Research code: GitHub | Writing: LaTeX

Algorithms: Reinforcement learning, knowledge transfer, constrained optimization, network simulation

## EXPERIENCE AND PROJECTS

### Research Assistant | Network Science Lab, UMass Amherst

Sep 2023 - Present

#### Cooperative 360-Degree Video Delivery | Python

- Built a Python simulation of servers delivering 360-degree video as small image regions (tiles), implementing content sharing and reuse under changing network conditions.
- Evaluated content reuse and streaming performance. Extending the platform with distributed learning to share experience across servers and adapt to new delivery conditions.

#### Adaptive Learning for Content Delivery | Python, PyTorch | [Code](#)

- Implemented a reinforcement learning system that assigns users to servers and selects content to cache or reuse, subject to latency and resource limits.
- Built Python/PyTorch training and knowledge-transfer components; evaluated 5 drone-mounted servers and 15 users across up to 10 environments. Released experiment scripts, saved results, and figure-generation tools.

#### Drone Task Scheduling and Network Resource Allocation | MATLAB | [Code](#)

- Implemented drone positioning, task assignment, and processing allocation using particle swarm optimization, dynamic programming, and lifelong learning. Published MATLAB training and online simulation modules, baselines, and saved results.
- Co-developed control methods for simulated satellite and ground networks, evaluating energy use, data freshness, and adaptation.

#### Semantic-Aware Task Assignment | Python | [Code](#)

- Built Python simulations assigning 60 computing tasks to 20 drone-mounted servers using Q-learning and particle swarm optimization. Compared greedy and random baselines; released parameter sweeps and CSV results.

## SELECTED PUBLICATIONS

First author: IEEE TMC, IEEE IoT Journal, IEEE VTC-Fall. Co-author: IEEE TWC.

## TEACHING AND PROFESSIONAL SERVICE

### Teaching Assistant, UMass Amherst | Sep 2023 - Present

- ENGIN 112 - Introduction to Electrical and Computer Engineering: Lead 30-40-student discussions five times every three weeks; independently design one-third of homework and quizzes.
- CS 250 - Introduction To Computation: Led weekly discussions for about 30 students and held weekly one-hour office hours.

Session Chair, IEEE VTC-Fall 2026 | Session 1E: Authentication and Privacy | Sep 2026.

Awards: NSF Student Travel Grant and IEEE VTS Travel Grant, IEEE VTC-Fall 2026.